AB-

- , GUE- :

(ORCID: 0009-0003-7299-0701)

22 · 2026

, « », ,
AB- : (–),
,
- . , ,
37 (18/19, 2026 .);
, — ,
(Markdown, HTML, PDF, DOCX),
.

. - , (`:monumental`)
AB- $\langle r \rangle = 0{,}585 \pm 0{,}026$
GUE 0,5992 — −2,4% (33), $L = 10, 20, 30, 50$
$0{,}548 \to 0{,}592 \to 0{,}605 \to 0{,}595$ (34). - ,
AB- 5000 $\zeta(s)$,
GUE, , $s \in [0; 3]$, –
($p = 6{,}7 \cdot 10^{-4}$; 35) —
. -
: $2\pi q$ (15, 24),
– $3{,}5 \cdot 10^{-15}$ 0,34 (25),
$C_1 = 2$ (20), $\alpha = 1/2$ 4 (17).
, $E_{\min} \propto 1/L$,
$R^2 = 0{,}9997$ (19), 20× $\alpha = 1/2$ (30)
- - $\sigma = 1/2$ (32).

: AB- — ,
GUE- - ,
, $\sigma = 1/2$,
- . ( ),
( ) (T), —
.

: – ; - ; ; GUE; – ;
; AB- ; ; ; TKNN; - – .

1.

1.1 –

1914 , - $\zeta(s)$
$\operatorname{Re} s = 1/2$. , 1958 , ( )
, – : $\hat{H}$,
$\{\gamma_n\}$, —
, $\operatorname{Re}\gamma_n = 1/2$ n. ,
: -
( ) .

, :
, .
, :
. — ,
,
. « » , , .
. 1972 ζ; ,
, ,
GUE ( )
. 1980- ,
1985 : (T)
GUE — , $R_2(0)$,
T. : ,
4–6 11 GUE $T \lesssim 4 \cdot 10^4$,
, .

1.2 – AB-

– ( ) ,
,
: , Φ, $\varphi_{\mathrm{AB}} = 2\pi\Phi/\Phi_0$,
$\Phi_0 = h/e$ — . : $t_{ij} \to t_{ij}e^{i\varphi_{ij}}$,
. — :
$\alpha = \Phi/\Phi_0$; α
« » (. 20).

AB- , — ,
$2\pi q$ (q — ,).
: « », — .

$$H_{ij} = -\exp(i\varphi_{ij}), \qquad \varphi_{ij} = 2\pi\alpha\, j + \sum_{k=1}^{N_v} q_k\big[\arg(\mathbf{r}_i - \mathbf{r}_k) - \arg(\mathbf{r}_j - \mathbf{r}_k)\big] + \varphi_{ij}^{\mathrm{str}},$$

j — ( ), N_v — , $\varphi_{ij}^{\mathrm{str}}$ — :
$-2\pi q$, , ( ), .
, $2\pi q$ —
(15, 24).

( ) , W: ,
. — $L \times L$, α, $\{q_k\}$,
N_v, W ( ). —
: ( ) , (
– , 25), ( ) :monumental (atan-),
q — GUE- .
: ,
— .

1.3

( )
.
, : **GOE** (, T-
), **GUE** (, T-) **GSE** ( ),
(,). ,
, 2.2; :
$\langle r\rangle \approx 0{,}386$ ( ), 0,531 (GOE), 0,600 (GUE). ,
T- GUE — « »,
AB- : $\alpha = 1/2$ - ,
$\tau(H) = \|H - H^{*}\|/\|H\| = 0{,}225 \neq 0$ (26), GUE,
$\tau = 0$ GOE (15, 16).

1.4 ,

— , AB- .
37 , : (1) $b(N)$
(1–14); (2) (15, 17, 20, 24–27); (3) GUE-
(16, 26, 29, 33, 34); (4) ζ (28, 35, 36); (5)
(19, 30–32, 37); (6) (21–23 8).
; (— , —
), « » 37 $16 \times 16 \to 32 \times 32$
5000 - .

, . (: , ,
–) $10^{-12} \ldots 10^{-16}$ PASS/FAIL. -
(GUE-) , : , PASS
$\langle r\rangle$ $0{,}5992 \pm 0{,}022$; WARN — «

, ». —
.

2.

2.1 -

- $\zeta(s) = \sum_{n\geq 1} n^{-s}$
$0 < \operatorname{Re} s < 1$. 50 000 ( ),
$T \in [14{,}13;\ 40433{,}7]$. T
– :

$$N(T) = \frac{T}{2\pi}\ln\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + O(\ln T),$$

$(T/2\pi)\ln T$ — « » ,
. **(unfolding)**:
$\delta_n = \gamma_{n+1}^{(u)} - \gamma_n^{(u)}$ 1 , .
$N(T)$; ,
, 10
(8,5% 10% — PASS).

: 50 000 (4, 6–10),
« » $T > T_{\min}$ (5) 1000/5000
(28, 35). GUE- , T;
(3.2).

2.2

( ) . ,
.

$\langle r\rangle$. E_{n-1}, E_n, E_{n+1}
$r_n = \min(\delta_n, \delta_{n+1})/\max(\delta_n, \delta_{n+1})$; ,
( ). ( ): 0,3863, GOE
0,5307, GUE 0,5996. ,
.

– **(KS)**. Wigner-
GUE: $D =$, n
5% $1{,}36/\sqrt{n}$. $n = 49999$ $D = 0{,}0061$ —
« » GUE; D n (4)
$D(T_{\min})$ (5).

$\Sigma^2(L)$ — L ( ).
$\Sigma^2 = L$; GUE $\Sigma^2 \approx (1/\pi^2)\ln(2\pi L) + \ldots$ — , .
$\Delta_3(L)$ — : ,

« » L; $L/15$, GUE $\approx (1/(2\pi^2))\ln(2\pi L)$.
9 L GUE- (13, 14).

$R_2(s)$ — s (
): GUE $R_2(s) = 1 - (\sin \pi s/(\pi s))^2$, « » $R_2(0) = 0$
; $R_2 \equiv 1$. $K(t)$ — - R_2, « + »
$K(t) = \min(t, 1)$ GUE; (35, 36).

– A^2 — KS, ; p-
- 2000 (11). , **f_GUE** — « GUE»:
$f_{\mathrm{GUE}} = (\Sigma^2_{\mathrm{P}} - \Sigma^2_{\mathrm{data}})/(\Sigma^2_{\mathrm{P}} - \Sigma^2_{\mathrm{GUE}})$ $L = 2$; $f_{\mathrm{GUE}} = 1$ — GUE,
0 — (29).

2.3 :

. $L_x \times L_y$; ; $H_{ij} = -\exp(i\varphi_{ij})$ (,).
j $2\pi\alpha j$. q_k $\mathbf{r}_k = (x_k, y_k)$ ( )
$(i \to j)$ $q_k[\arg(\mathbf{r}_i - \mathbf{r}_k) - \arg(\mathbf{r}_j - \mathbf{r}_k)]$, — $-2\pi q_k$
, ; , atan- .
: $2\pi\alpha + 2\pi q_k \delta_k$ ($2\pi\cdot$), δ_k —
, . 224 16×16: | |
$1{,}5 \cdot 10^{-14}$ (24).

W $\varepsilon_i \in [-W/2; W/2]$
(: —). ,
(–) : q ,
25 $3{,}5 \cdot 10^{-15}$.

— . ,
$\langle r \rangle$ (33: 0,442 0,585
). $q\,(2 - N_x) \in \mathbb{Z}$ —
; q , $q = 0{,}3$ 16×16
, ( ).

2.4

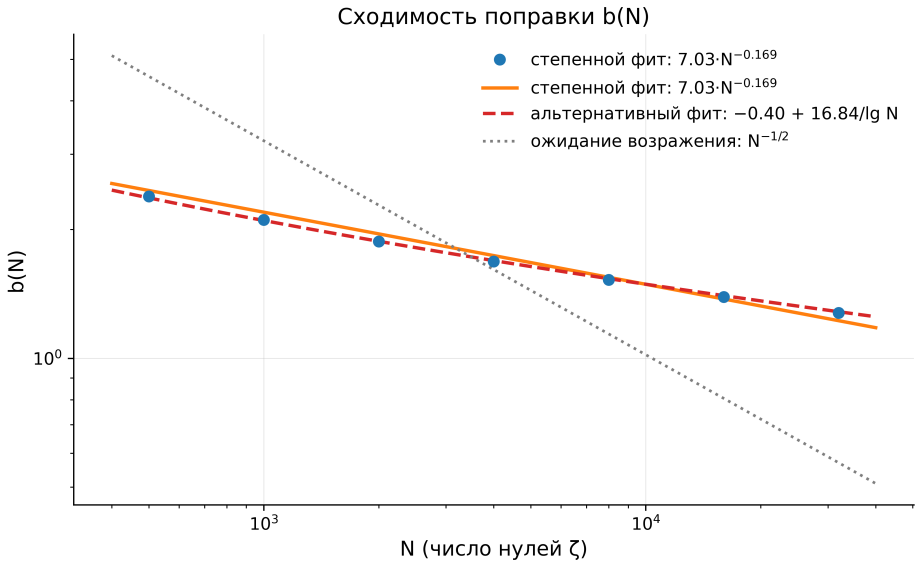
`results/run_< - >/` : —
( ), ,
Markdown, HTML, PDF ( ) DOCX, PNG (600 dpi),
SVG PDF GIF- . ,
: , , , .
— `julia ab_cloud_v19.jl --test <N> --no-two-pass`;
— `--test all`; — `--quick` (16×16→32×32, 5000 ,).
: (v19 ,
. A).

3. 1: $b(N)$ ζ (1–14)

3.1 $b(N)$

$b(N)$: ,
.
$\sim N^{-1/2}$; , . :
6 $b(N)$ — ,
1 , . 3, 7, 9
— .

3.2

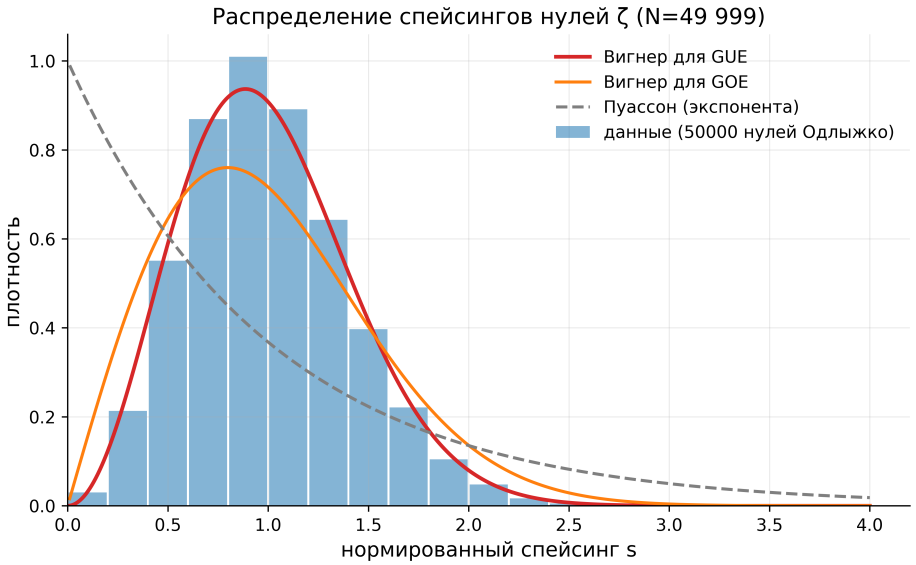


. 1: **. 1.** $b(N)$: ($1/\ln N$).
$N^{-1/2}$, , .

- (3) $N \in [50; 25600]$ $b(N) \approx 7{,}03\,N^{-0{,}1685}$, $R^2 = 0{,}9895$, $0{,}0043$. $N = 32000$ (7) -
$-0{,}1504$ 95%- $[-0{,}1594; -0{,}1414]$, $R^2 = 0{,}9954$ —
. (9, 1000) $-0{,}1746$ 95%- -
$[-0{,}1895; -0{,}1552]$. , : $b(N) \approx -0{,}40 + 16{,}84/\ln N$ ($R^2 = 0{,}9999$ $0{,}9954$
7). , — , ;
. 1 3 , N .
(8) (3 5,8 ;
1 0,68) — « » , ;
.

: $N \le 5 \cdot 10^4$ $b(N)$ (2,39 $N = 500$ 1,28 $N = 32000$) — - ,
«GUE », $b(N)$, .

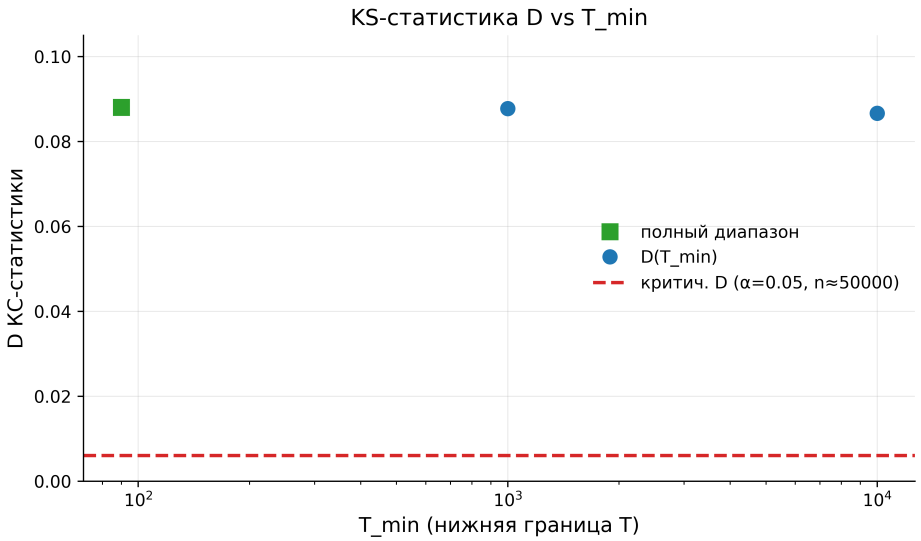
3.3 2: KS, ² A² GUE —



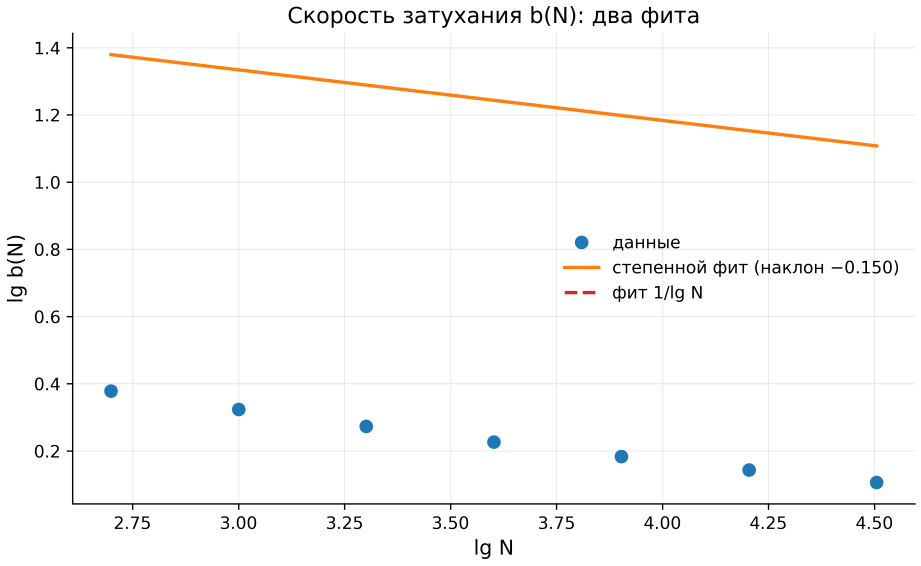
. 2: **. 2.** 49 999 ζ (GUE, GOE)
: « », .

4, 5, 6, 11 .
49 999 KS- GUE $D = 0{,}0881$ 0,0061 — 14 , H_0 (GUE) $p \approx 10^{-300}$. -
150/300/600 $\chi^2 = 6170/6276/6384$ 149/295/550 — . – 5000 GUE-
$A^2 = 117{,}7$ - 95- 16,6 ($p \approx 0$ 2000).

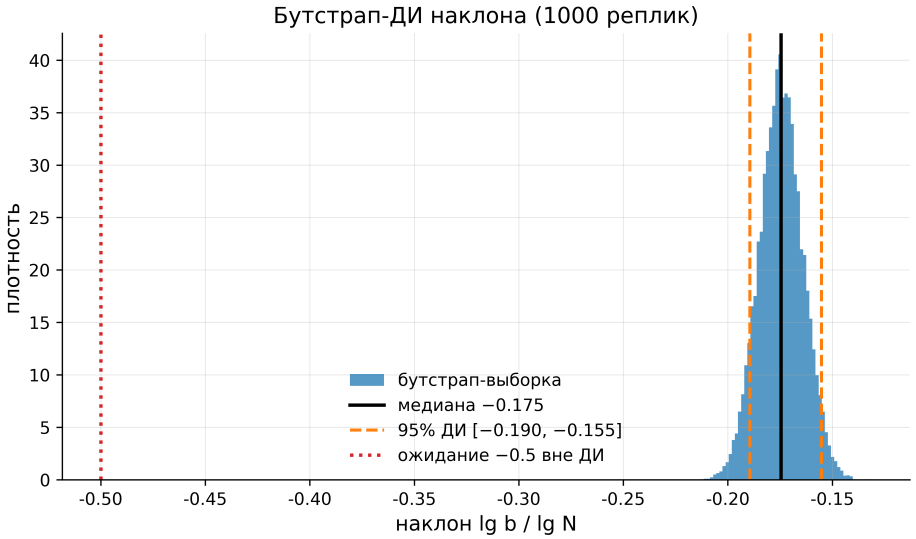
GUE- — . GUE-
: $T \gg 10^6$, . (1985)
, : $T \le 4{\cdot}10^4$
$D \sim 0{,}08$. 5 (: $D = 0{,}0878 \to 0{,}0866$
$T_{\min}$: $1000 \to 10000$ — ,) , , 12:
KS GUE- 2000×2000 $D = 0{,}0406$,
$p = 0{,}139$ — « GUE- »
. : GUE- ,
GUE- — GUE- .



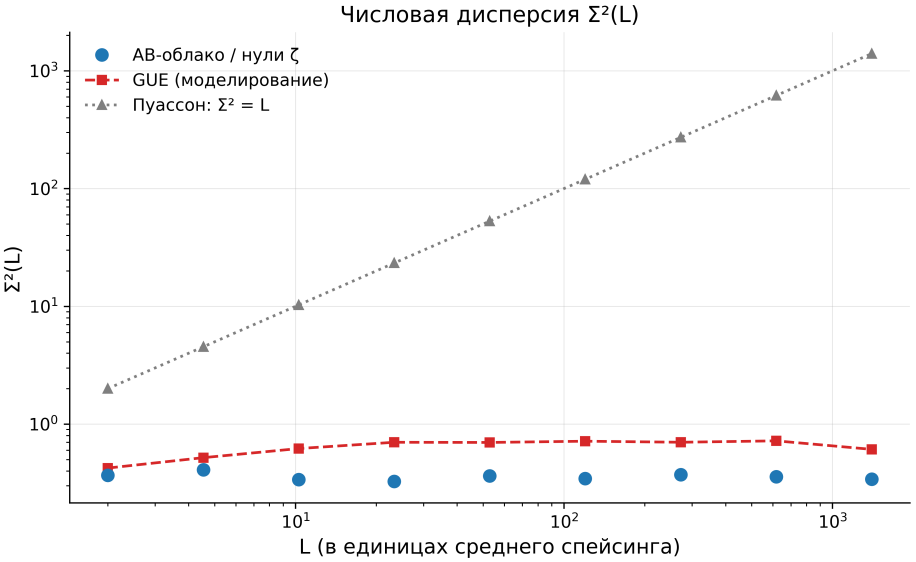
. 3: **. 3.** KS- D $T_{\min}$: — GUE.



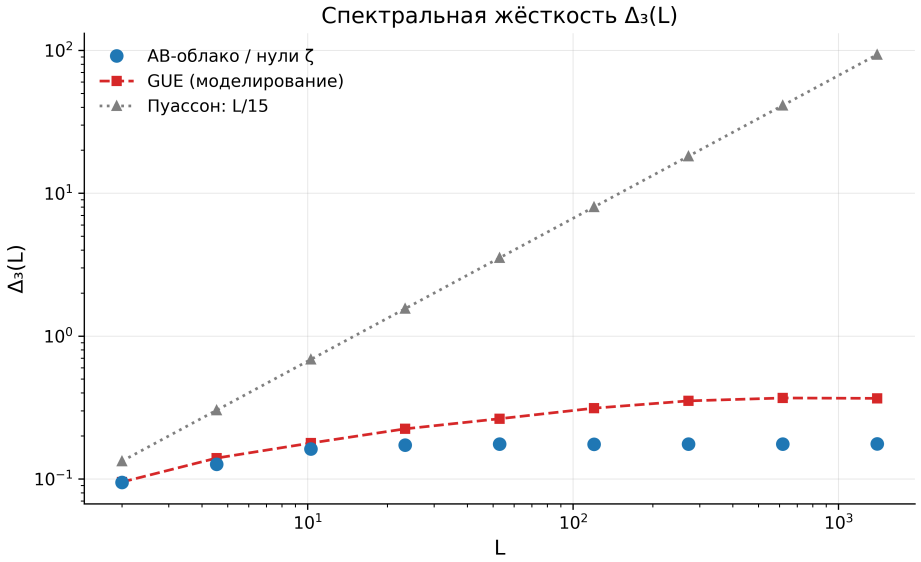
. 4: **. 4.** $b(N)$ - : $1/\ln N$; .



. 5: **. 5.** - (1000): 95% $[-0{,}190; -0{,}155]$; $-0{,}5$.



. 6: **. 6.** $\Sigma^2(L)$: ( ) GUE — 9/9 GUE.



. 7: **. 7.** $\Delta_3(L)$: GUE L.

3.4 : GUE, GUE-

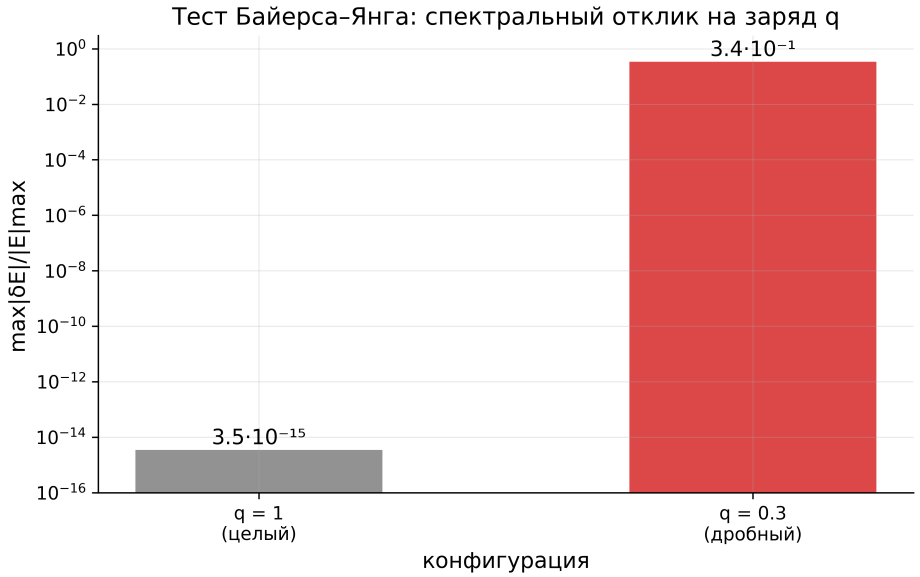
$\Sigma^2(L)$ $\Delta_3(L)$ (13, 14). $L \in [2; 1400]$ $\Sigma^2 \approx 0{,}33$–$0{,}41$ — , 1400, GUE- — 0,61–0,72. GUE, , 9/9 ; ( ) GUE L — Δ_3 GUE- 0,99 $L = 2$ 0,48 $L = 1400$. ζ ( ), — : (,) Δ_3 Σ^2, .

4. 2: AB- (15, 17, 20, 24–27)

4.1 :

, . 15 16×16 $q = \pm 1$: $\|H - H^\dagger\|/\|H\| = 0$ ( ); — $\tau(H) = \|H - H^*\|/\|H\| = 0$, (GOE,); — 10^{-14} . 24 225 : 1/1 OK, 224/224 OK, $1{,}47 \cdot 10^{-14}$. — : , .

4.2 — : , —



. 8: **. 8.** — : $(3{,}5 \cdot 10^{-15})$, — $(0{,}34)$; .

— (1961) :
— « » ( ) ,
. 25 : $q = \pm 1$ 16×16
$\alpha = 0$ $3{,}49 \cdot 10^{-15}$ —
, . $q = 0{,}3$ $0{,}342$ —
$4\pi \cdot 0{,}3 = 1{,}2\pi \not\equiv 0 \pmod{2\pi}$. ,
: GUE-
$\alpha \neq 0$; `monumental` q, (5).
, « », .

4.3 AIII

$\alpha = 1/2$: $\alpha \to 1/\alpha$
, $\alpha = 1/2$. 17 : $\alpha = 1/2$
4 (L —), E→−E-
— $1{,}4 \cdot 10^{-15}$. (27):
Γ ( ) $\Gamma H \Gamma = -H$ α ;
$W = 0$ 0 ( ), $W = 4$ — $5{,}9 \cdot 10^{-3}$ (, ,
), — 0: .
17: $\alpha = 1/3$ $E = 0$ (0),
$\alpha \leftrightarrow 1/\alpha$ (0,30) — ( ) ;
, .

4.4 : C = 2

20 – – 6×6 $\alpha = 1/2$ ( ): 8×8 , $C_1 = 2$. $|C_1| = 1$; $\alpha = 1/2$, « » $C_1 = \pm1$ — $|C_1| = 2$. : $C_1 \neq 0$ — , , (« » , . 20).

4.5 T-

26 : 16×16, $q = 1$ - $q(2 - N_x) = -14 \in \mathbb{Z}$, +x. : 255 — OK; — OK; $\tau(H) = 0{,}2252$ — T- (GUE), $\langle r\rangle = 0{,}5857$ — GUE $0{,}5992 \pm 0{,}022$- . — : + T + GUE- .

5. 3: GUE- AB- (16, 26, 29, 33, 34)

5.1 : GOE GUE

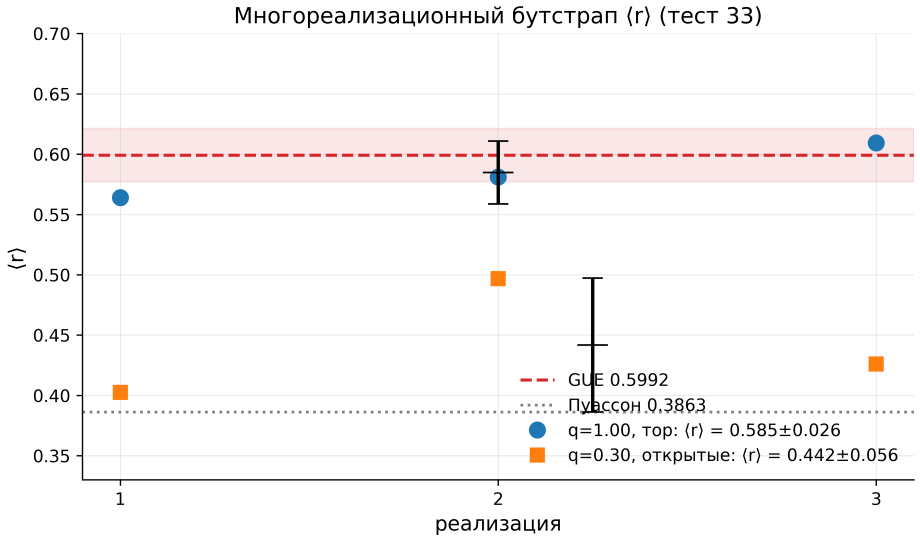
16 ( ) $N_v = 2$, $q = \pm1$, $\alpha = 1/2$ $\langle r\rangle = 0{,}5133 \pm 0{,}0353$ — GOE (0,531) GUE (0,600), - p- 0,00 GUE- 30 256×256. 4.2: , , « » GOE — , T- . 16 33: $\langle r\rangle$ 0,585. : — ; , .

5.2 (33)

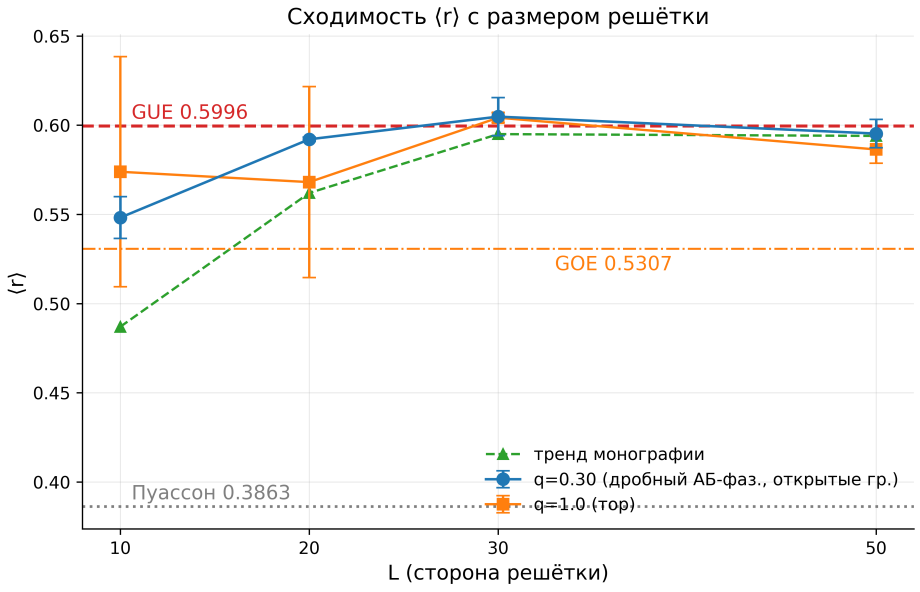
. : 16×16, $q = \pm1$, $\alpha = 1/2$, $W = 4$, , (,). : $\langle r\rangle = 0{,}5639,\ 0{,}5811,\ 0{,}6094$; $\langle r\rangle = 0{,}5848 \pm 0{,}0260$, 95% $[0{,}5588; 0{,}6108]$, GUE- 0,5992 $-0{,}0144$ $(-2{,}4\%)$ — , PASS. $q = \pm0{,}3$: 0,4025, 0,4967, 0,4260 $\Rightarrow$ $0{,}4417 \pm 0{,}0555$ — , ( ) . (0,585 0,442) — (. 11).

5.3 (34)

$\langle r\rangle(L)$ $L = 10, 20, 30, 50$ (, $W = 4$): $q = 0{,}3$ ( ) — $0{,}5482 \to 0{,}5921 \to 0{,}6048 \to 0{,}5953$; $q = 1$ ( )



. 9: **. 9.** 33: $\langle r \rangle$ $q = 1$ ( ) $q = 0{,}3$ ( ) GUE-
.



. 10: **. 10.** 34: $\langle r \rangle$; GUE- $\approx 0{,}60$.

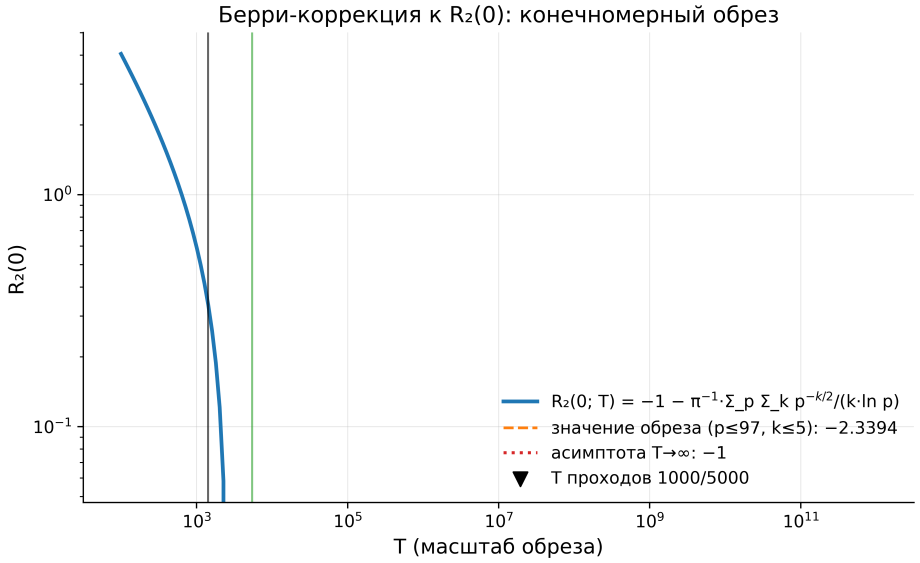
— 0,5739 → 0,5681 → 0,6042 → 0,5864. $L \geq 30$ 0,59–0,605
GUE 0,5996;
(0,487 → 0,562 → 0,595 → 0,594) — $L = 10$
(L), .
« » : $\langle r \rangle$ « » , GUE- ;
v13 .

5.4 f_GUE (29)

29 : 16×16, $N_v = 2$, $q = \pm 1$, $\alpha = 1/2$, $W = 4$
$\Sigma^2(L{=}2) = 0{,}8377$ $f_{\mathrm{GUE}} = 0{,}648$ PASS 0,90 — WARN.
33: Σ^2 L « » , ;
0,90 $\geq 100 \times 100$. 5 $K(t)$:
, , .
, , .

6. 4: (28, 35, 36)

6.1 (28)

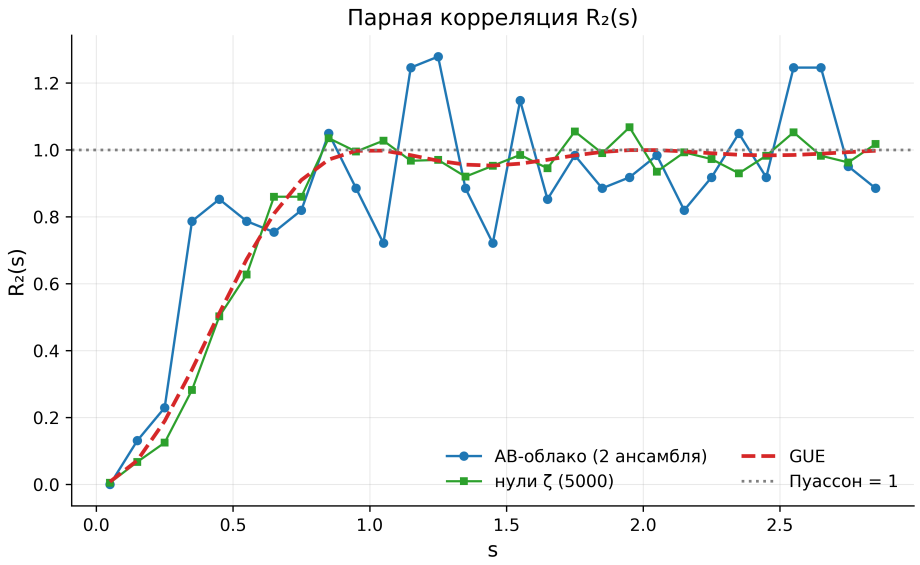


. 11: **. 11.** - $R_2(0;T)$: -1;
— (1000/5000).

28 — - $R_2(0)$ — v18/v19 :
1000 ($T_{\max} \approx 1419$) 5000 ($T_{\max} \approx 5448$).
$\sum_p \sum_{k \leq 5} p^{-k/2}/(k \ln p) = 4{,}2079$ $R_2(0) = -2{,}3394$;

$\sim T^{-1/2}$ 0,0019 ($T = 1419$) 0,0009 ($T = 5448$).
(PASS,)
- : $R_2(0)$ « »
$-1{,}007$. : ,
« » .

6.2 : AB- -5000 (35)

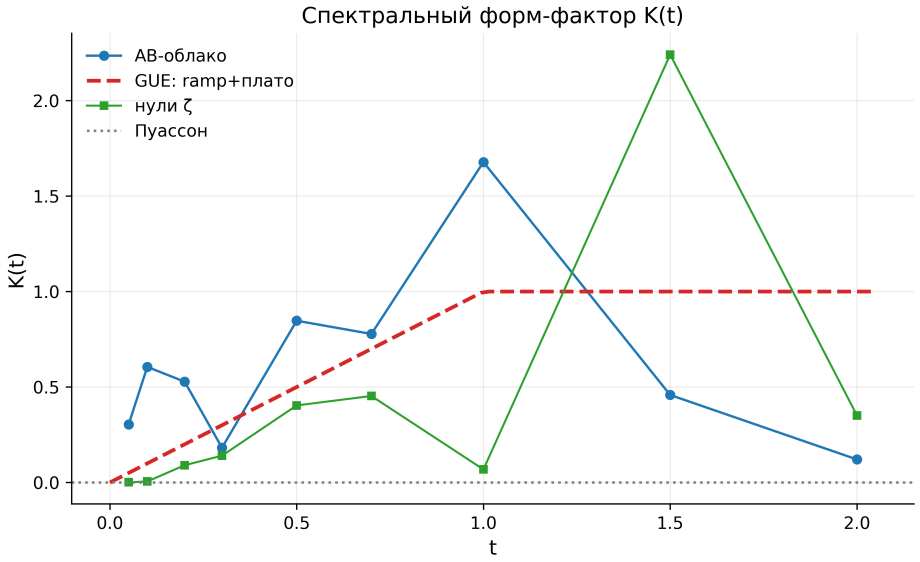


. 12: **. 12.** $R_2(s)$: AB- ζ, GUE —
.

: AB- (2 , 304)
5000 (4999 ,). : KS $D = 0{,}1173$, $p = 6{,}7 \cdot 10^{-4}$ — ;
$\langle |R_2^{\mathrm{AB}}(s) - R_2^{\zeta}(s)| \rangle = 0{,}147$ 0,10. $R_2(s)$ (29 s
0,05 2,85; . 8) : — $R_2^{\mathrm{AB}}(0{,}05) = 0{,}00$,
$R_2^{\zeta}(0{,}05) = 0{,}005$; GUE-
« GUE, » ($d_{\mathrm{GUE}} = 0{,}140 < d_{\mathrm{Pois}} = 0{,}227$). :
AB- (, $s \approx 1$,
), — 304 R_2, KS- .
« » « » , .

6.3 K(t) (36)

— :
. 306 RMS- GUE- 0,93 (PASS
0,30) 0,018 (0,50) — WARN ;



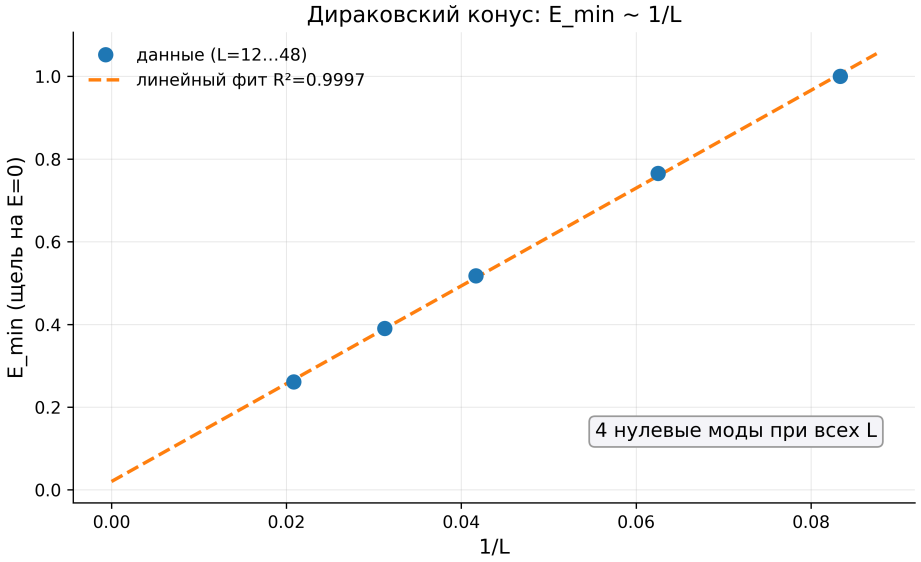
. 13: **. 13.** $K(t)$: + GUE, AB- ζ;
.

(. 9) $t \approx 1$, , .
(5000) RMS GUE 0,86 —
: , .
.

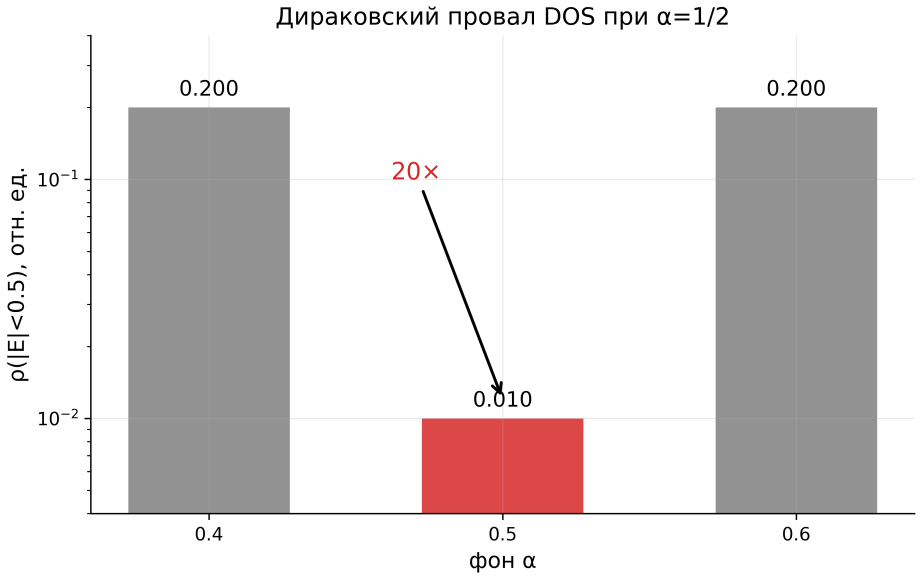
7. 5: (19, 30–32, 37)

7.1 (19, 31)

AB- $\alpha = 1/2$: .
(19) $L = 12, 16, 24, 32, 48$: $E_{\min}$ (
) $E_{\min} = 0{,}0202 + 11{,}826/L$ $R^2 = 0{,}9997$ —
$1/L$. : $(k_{\min} = 2\pi/L)$,
$v_F = b/2\pi \approx 1{,}88$ (b);
$(k_{\min} = \pi/L)$ — $v_F = b/\pi \approx 3{,}76$. $\sim 0{,}125$
; , —
, . 31 (
) $R^2 = 0{,}03$:
L, —
.



. 14: **. 14.** : $E_{\min} = 0{,}02 + 11{,}83/L$, $R^2 = 0{,}9997$ — $1/L$.

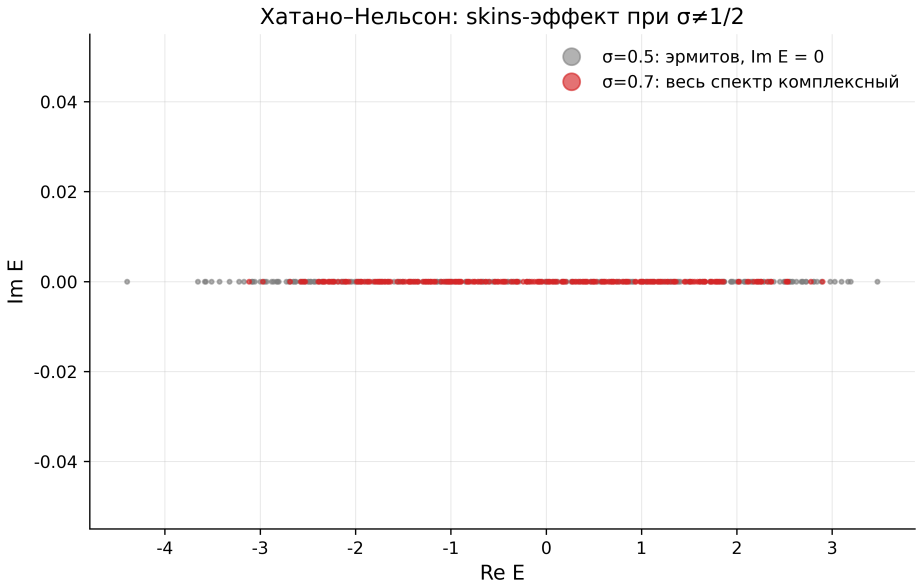


. 15: **. 15.** DOS $\alpha = 1/2$: $20\times$ ( ).

7.2 (30)

$\alpha = 1/2$: $E = 0$
$(\rho(E) \sim |E|$), $\alpha \neq 1/2$. 30
$|E| < 0{,}5$ 20×20 : $\rho(\alpha = 0{,}4) = 0{,}200$,
$\rho(\alpha = 0{,}5) = 0{,}010$, $\rho(\alpha = 0{,}6) = 0{,}200$ — 20× .
— « » , $C_1 = +1$ $C_1 = -1$;
( ) —
, , .

7.3 – : $\sigma = 1/2$ (32)



. 16: **. 16.** – : $\sigma = 0{,}5$ — ; $\sigma = 0{,}7$ —
(-).

σ « » ; $\sigma = 1/2$. 32
$\sigma = 0{,}5$ $\sigma = 0{,}7$ 16×16: $\sigma = 0{,}5$ $|\mathrm{Im}\, E| = 0$ (
); $\sigma = 0{,}7$ 256 , $|\mathrm{Im}\, E| = 0{,}623$,
- – $g = (\sigma - 1/2)\ln|\gamma| = 0{,}322$. : $\sigma = 1/2$
(GUE) ( ) ;
(-) **,** $\langle r \rangle$ (« »
). $\sigma = 1/2$, ,
(. 13, 16). ,
« » — .

7.4 (37)

: ? 37 , $\langle r \rangle$
256 (0,0015 0,02)
7,9991 8 (χ^2- : $z = -0{,}234$, $|z| < 3$).
:
($\langle r \rangle_{\text{raw}} = 0{,}447$), 15 $\langle r \rangle_{\text{local}} = 0{,}549$ —
, 5.4.

8. 6: (21–23)

— . : ( ) ,
(21–23); ( ) , , ;
( ) , . — .

8.1 γ^*: 90° (21)

$\delta_C = \pi/7 = 0{,}448799$ — . 8.2 — $b_2(K3) = 22$ (
K3-). 21 : $a_C(\gamma^*) =$
$\delta_C^5 / b_2(K3) = 8{,}276 \cdot 10^{-4}$ - $b_C(\gamma^*) = 1 - \cos(2\delta_C) = 2\sin^2\delta_C =$
0,376510; . $\gamma^* = a_C + i b_C$
$|\gamma^*| = 0{,}3765$ 89,874° — $\pi/2$ 0,126° ($2{,}2 \cdot 10^{-3}$).
, 21: $b_C \gg a_C$, γ^* 90° — «
90° »: , ( )
. – — ( ):
, .

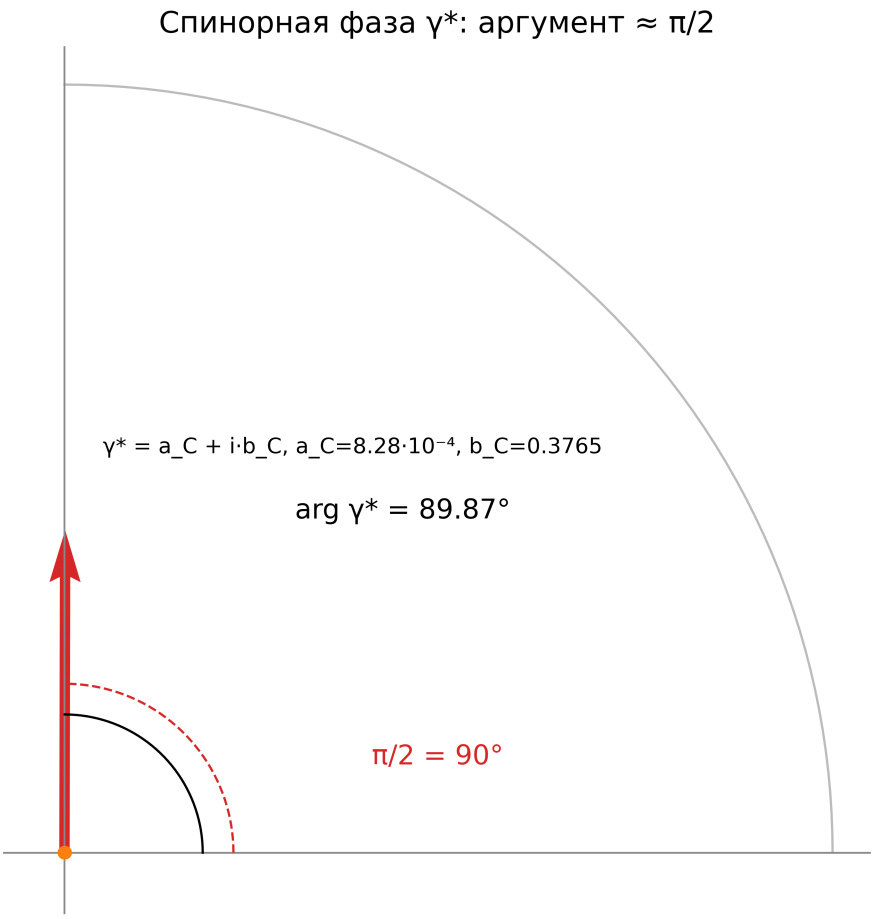
8.2 - $\Phi_{AB} = \pi/7$ (22)

: $q/e = 1/14$, , –
$\Phi_{AB} = 2\pi/14 = \pi/7 = 0{,}448799$ $= 25{,}714°$. 22
(0,00). $\delta_C = \pi/7$ — Φ_{AB}, $q/e = 1/14$
— , . , $\pi/7$
$b_7 = (-1 + i\sqrt{7})/2$ PSL(2,7) (8.5):
, — , v21.

8.3 CF (23)

$\lambda_{\text{DSI}} = b_2(K3) = 22$ 23

$$CF = \exp\!\left(\frac{2\pi i}{\lambda_{\text{DSI}}}\right) \cdot \exp\!\left(i\,\beta_{\text{imag}}\,\ln\lambda_{\text{DSI}}\right), \qquad \beta_{\text{imag}} = 2 + \frac{1}{2\sin^2(\pi/7)} - 2\cos\frac{\pi}{7} = 2{,}854033,$$



. 17: **. 17.** $\gamma^* = a_C + ib_C$: 89,87° 90°; b_C $\pi/2$.

$CF = -0{,}9501 + 0{,}3119\,i$, $|CF| = 1$. —
« »:
.
: tex- $CF_{\text{hard}} = 0{,}9786 + 0{,}0390\,i$
(0,9794, $c_{AB} = 1 - |CF_{\text{hard}}| = 0{,}0206$) — , α_s ,
— ( ) ; c_{AB},
.

8.4 T- GUE:

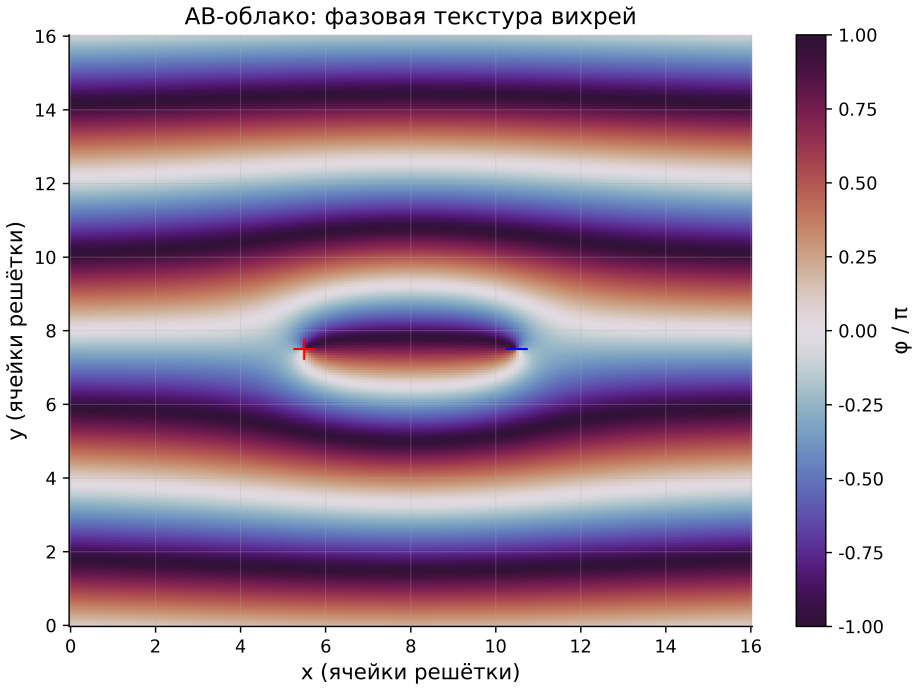
3–5 : GOE T- , GUE .
AB- : — ,
$\tau = 0$, GOE- (15, 16); — $\tau \approx 0{,}225$, GUE-
(26, 33). : **T-** —
, **(** **)**
. : ,
, — «
$\langle r \rangle \to 0{,}6$» .

8.5 - , PSL(2,7) Monster ( )

21 . **-** : « » —
, , ; ,
, . : χ_2 196883
PSL(2,7) ($2^{3+6+12+18}.(L_3(2) \times 3S_6)$)
$1456 \cdot \chi_1 + 3334 \cdot \chi_3 + 3334 \cdot \chi_3' + 6784 \cdot \chi_6 + 8081 \cdot \chi_7 + 9770 \cdot \chi_8$
(196891, 0,004%);
. class fusion GAP CTblLib — ,
.

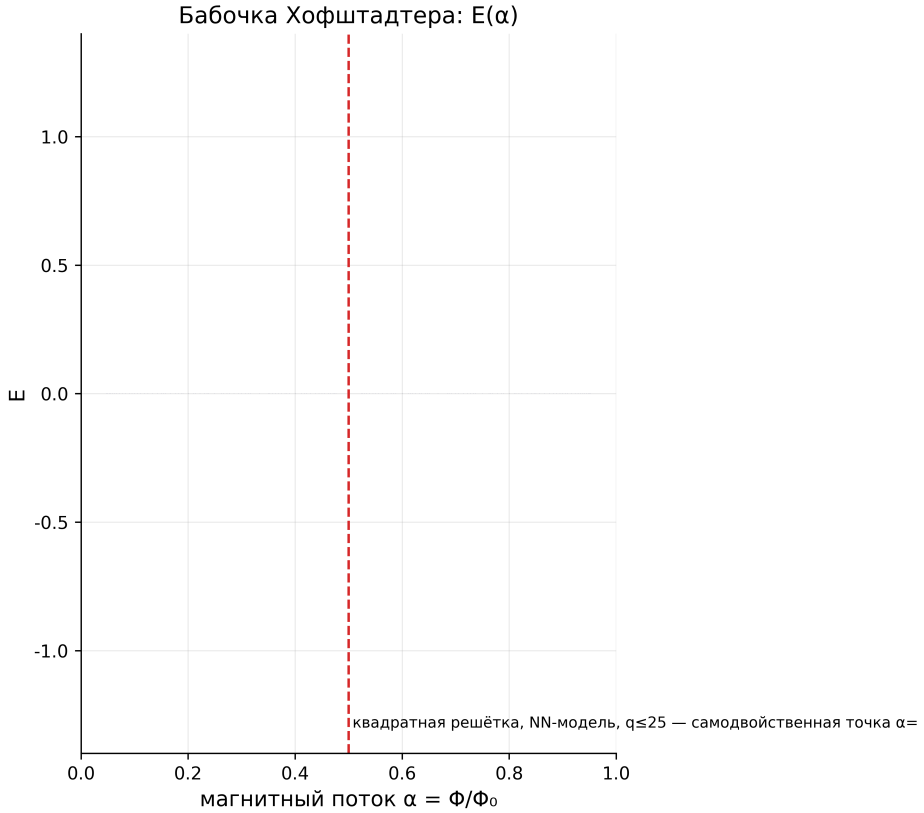
9. :

. (1) AB- $\alpha = 1/2$ GUE- ,
: $\langle r \rangle : 0{,}548 \to 0{,}595$ $L : 10 \to 50$ (33, 34). (2) — GUE
(4–14); « GUE» , «GUE
» — (12). (3) R_2 (
35) — . (4) $\sigma = 1/2$ —
, (17, 30, 32) —
KS- . (5)
(, 19), — (– , 25). ,
« », **,**
, : , (,),
(,) ($T \to \infty$,
R_2).


АВ-облако: фазовая текстура вихрей
y (ячейки решётки)
x (ячейки решётки)
φ / π
0
2
4
6
8
10
12
14
16
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00

. 18: **. 18.** $q = \pm 1$: « » ,

.



. 19: **. 19.** $E(\alpha)$; — $\alpha = 1/2$,

.

10. : «PASS»

19 — , .
: v18 PDF- (`String[]`,) `MethodError` ; , WARN, — « » , . : 37 WARN, PDF- , . (FIX-R1…R5 A) : (1) PDF- ; (2) ; (3) 28 ( ) 18 ($\alpha = 1/3$,) ; (4) 5000 16×16→32×32.
: **weakest-** , « » , .

: — ; ; `julia ab_cloud_v19.jl --test N --no-two-pass; --quick;` , ( ).

11.

1. ( ): 0; 10^{-14}; – $3{,}5 \cdot 10^{-15}$; — 4 E→−E 10^{-15}; 0,0 $W = 0$. — , .
2. **GUE-** ( ): $\langle r \rangle = 0{,}5848 \pm 0{,}0260$ GUE 0,5992 (−2,4%); $L = 10 \to 50$ $0{,}595 \pm 0{,}008$; L GUE 9/9 . T- , , — .
3. ( ): $R_2(s)$ (, ; GUE,), 304 ; GUE.
4. : $C_1 = 2$ (IQHE), $R^2 = 0{,}9997$, DOS 20×, - $\sigma \neq 1/2$; $\sigma = 1/2$ (+ +).
5. : γ^*, $\Phi_{AB} = \pi/7$, CF $\pi/7$- ; (– ,) — , .
6. : v19 ; ; ( ) .

12.

1. $b(N)$: ? ($R^2 = 0{,}9999$), N ; $N \sim 10^6$ (2M).

2. R_2: AB- $\geq 10^4$ (× ×) — KS -5000 .
3. $K(t)$: ; PASS (RMS $<$ 0,30, corr $>$ 0,50) , , ≥ 3000 .
4. **3D**: (3D-) — GUE- ; , 37- .
5. **196883 PSL(2,7)**: class fusion GAP CTblLib; .
6. **-** : ; - .

A. :

(`ab_cloud_v19.jl`, Julia 1.10, — PNG- pHYs, PDF, GIF89a LZW, DOCX stored-ZIP) `code/`; , `f` — . .

: (1) `Config` ; (2) — 50 000 + 2M ; (3) (, KS, ^{2}, AD, , Σ^2, Δ_3, R_2, $K(t)$); (4) AB- `build_ab_cloud_hamiltonian` (`:dirac`, `:monumental`) `plaquette_flux`; (5) 37 - `test*`; (6) `run_test_two_pass`; (7) v16/v19 (`rep_*`, md/html/pdf/docx/svg/png/gif, index.html); (8) (22 -) 3D- (30); (9) CLI.

— (; —):

```
# _ij = 2 · ·j + Σ_k q_k·[arg(r_i−r_k) − arg(r_j−r_k)] + ^   _ij
#     (i→j)       H_ij = −exp(i _ij);            k          2 ·q_k
function build_ab_cloud_hamiltonian(cfg::ABCloudConfig)
    N = cfg.Nx * cfg.Ny
    H = zeros(ComplexF64, N, N)
    for iy in 0:(cfg.Ny-1), ix in 0:(cfg.Nx-1)
        i = iy * cfg.Nx + ix
        for (dx, dy) in ((1,0),(0,1))            #          ,
            jx, jy = (ix+dx) % cfg.Nx, (iy+dy) % cfg.Ny
            j = jy * cfg.Nx + jx
              = 2  * cfg.alpha * jy              #            (      )
            for v in cfg.vortices                #        (            )
                 += vortex_phase(v, ix, iy, jx, jy, cfg.model)
            end
            H[i+1, j+1] -= exp(im *  )
            H[j+1, i+1] -= exp(-im *  )          #
        end
        H[i+1, i+1] += disorder_epsilon(ix, iy, cfg)   #   ~ U(-W/2, W/2),
```

```
    end
    return H
end
```

( ): $2\pi q \pmod{2\pi}$, — 0; `plaquette_flux(H, Nx, Ny, ix, iy)` 15/24/26.

: `log_result` ; `run_test_two_pass` , `finally` ( ); `rep_finish!` , (FIX-R2) .

v19 ( ): FIX-R1 — PDF- (MethodError,); FIX-R2 — try/catch `_write_test_report!`; FIX-R3/3b — `last_R2/berry_ref` 28; FIX-R4 — 6 {1/2,1/3} 18; FIX-R5/5b — : 5000, GIF, CLI- `--quick`.

B.

```
#       (       , ~       ):
julia ab_cloud_v19.jl --test all
#         :
julia ab_cloud_v19.jl --test 33 --no-two-pass
#           16×16 → 32×32 (   5000, ~   ):
julia ab_cloud_v19.jl --quick
#       :
julia ab_cloud_v19.jl --lang ru --quick
#    : results/run_<  >/test_NN_<slug>/{report.md,html,pdf,docx, plots/, logs/}
```

v19 `run_20260902_134759` 2026-09-02 (37 , two-pass 72×72 → 96×96 HARDCORE- , pass-3 19/29/31; Julia 1.12.0; 50 000). : **32 PASS / 5 WARN** (4, 5, 6, 11, 12, 28, 35 — WARN- « », KS- GUE- ;); **HARDCORE-** — **PASS** (77). — (report.md/pdf/docx/html), - (logs/), FINAL_REPORT index.html — : `results/run_20260902_134759/`. 64 `verification/spinor64` (D). v18 (`results/verification_run_v18_37tests_2026-08-28.txt`) .

C.

— 600 dpi, PNG, ; `monographs/<lang>/figures/`.
. 1 — $b(N)$ ( ); . 2 — ; . 3 — - ; . 4 — - 95% ; . 5 — KS $D(T_{\min})$; . 6 — $\Sigma^2(L)$; . 7 — $\Delta_3(L)$; . 8 — $R_2(s)$: AB- , GUE, ; . 9 — $K(t)$; . 10 — $\langle r\rangle(L)$ $q = 0{,}3$ $q = 1$; . 11 — $\langle r\rangle$ (33); . 12 — $E_{\min}(1/L)$; . 13 — DOS; . 14 — – ( ); . 15 — - $R_2(0;T)$; . 16 — – , ; . 17 — γ^*; . 18 — ; . 19 — $\alpha = 1/2$.

D. 64 (v22.1, spinor64)

`verification/spinor64` (Python/NumPy, : `python3 verification/spinor64/run_spinor64.py`) 64 $\varepsilon \in \mathbb{F}_2^6$

E1 — ({3,7}). {3,7} (56 , 84 , 24 , Aut = PSL(2,7) 168). 64 - (– – , :). PSL(2,7)- **28 / 21 / 7 / 7 / 1**; 28 — (Arf=1) — , PSL(2,7) 28 (–). - : 64 $\mathbf{8.9 \cdot 10^{1}}$; $7.1 \cdot 10^{1}$; : 2 ( ) / 3 ( ) / 7 ( ). : , PSL(2,7) (. . 3.2.3 v21.1). v21 $\mathrm{Arf}(\varepsilon) = \varepsilon_1\varepsilon_2 + \varepsilon_3\varepsilon_4 + \varepsilon_5\varepsilon_6$).

E2 — (AB-). $L = 44$, $\alpha = 1/2$, $N_v = 54$ $q = +1$ ($N_v = 25$ 30×30), $W = 0$, `:monumental` (atan- , , 0.5), - — $\varphi_x = \pi(\varepsilon_1 + \varepsilon_3 + \varepsilon_5)$, $\varphi_y = \pi(\varepsilon_2 + \varepsilon_4 + \varepsilon_6)$; 5 , 0.6. — Monte-Carlo GUE 100 1936×1936 (16): $\langle r\rangle = 0.6013$, 95% $[0.5847; 0.6140]$. : **64** : $\langle r\rangle = 0.5984 \pm 0.0035$ ($0.5935 \ldots 0.6027$, min MC p = 0.36). GUE- — GUE, 4.1, — AB- , .

38 (+ , LAPACK) 10 : `verification/{cpp,java,rust,go,fortran,haskell,r,matlab,julia,javascript}/spinor38/`.